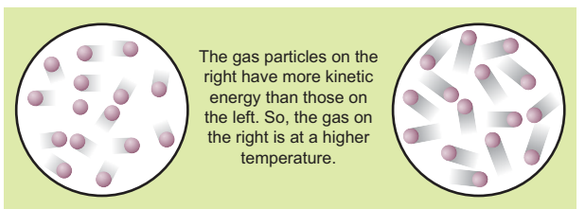
Introduction



You shiver when it is cold outside and sweat when it is hot outside, but how can you measure those weather temperatures? Temperature is involved in many aspects of our daily lives, including our own bodies and health; the weather; and how hot the stove must be in order to cook food.

The measurement of warmness or coldness of a substance is known as its temperature.

It is a measure of the average kinetic energy of the particles in an object. Temperature is related to how fast the atoms within a substance are moving.



1.2 Temperature Units:

There are three units which are used to measure the temperature: Degree Celsius, Fahrenheit and Kelvin.

Degree Celsius: Celsius is written as°C and read as degree. For example 20°C; it is read as twenty degree Celsius. Celsius is called as Centigrade as well.

Fahrenheit: Fahrenheit is written as °F for example 25°F; it is read as twenty five degree Fahrenheit.

Kelvin: Kelvin is written as K. For example 100K; it is read as hundred Kelvin.

The SI unit of temperature is kelvin (K).

1.3 Measuring Temperature

The temperature of the object is well approximated with the kinetic energy of the substances. The high temperature means that the molecules within the object are moving at a faster rate.

But the question arises, how to measure it? Molecules in any substance are very small to analyze and calculate its movement (Kinetic energy) in order to measure its temperature. You must use an indirect method to measure the kinetic energy of the molecules of a substance.

We studied that solids expands when heat is supplied to it. Like solid substances, liquids are also affected by heat. To know this let us do the activity 1.

In a thermometer, when liquid gets heat, it expands and when it is cooled down, it contracts. It is used to measure temperature.

Like solid and liquid objects, the effect of heat is also observed on gaseous objects.

1.4 Thermometer:

Thermometer is the most common instrument to measure temperature.

There are various kinds of thermometers. Some of them are like glass tubes which look thin and are filled with some kind of liquid.

Why Mercury or Alcohol is used in Thermometer?

Mostly Alcohol and Mercury are used in thermometers as they remain in liquid form even with a change of temperature in them. A small change in the temperature causes change in volume of a liquid. We measure this temperature by measuring expansion of a liquid in thermometer.

VII Std Science Term-2 EM Unit 1.indd 2 11/11/2022 18:02:55